

Retail Customer Behavior & Sales Analytics

1. Executive Summary

This project analyzes a comprehensive consumer behavior dataset from a leading retail company to uncover deep insights into shopping patterns. The primary objective is to provide data-driven recommendations that improve sales performance, enhance customer satisfaction, and foster long-term brand loyalty. The management team has identified shifting purchasing trends across various demographics, product categories, and sales channels (online vs. offline). This analysis focuses on isolating the core factors that drive consumer decision-making and repeat purchases, including sensitivity to discounts, product reviews, seasonal trends, and payment preferences.

2. Business Problem Statement

The entire data querying and visualization pipeline is designed to address the following overarching strategic question:

"How can the company leverage consumer shopping data to identify actionable trends, improve customer engagement, and holistically optimize marketing and product strategies?"

3. Project Deliverables & Architecture

To solve this business problem, the project is structured as an end-to-end data pipeline, moving from raw data processing to a fully interactive business intelligence solution. The technical components within this repository include:

- ***Data Preparation & Transformation (Python / ETL):*** Cleaning and transforming the raw dataset to establish a structured format optimized for deep-dive analysis.
- ***Data Analysis & Modeling (Microsoft SQL Server):*** Organizing the data into a relational schema, simulating real-world business transactions, and executing advanced T-SQL queries (including CTEs, Window Functions, and Subqueries) to extract insights regarding customer segmentation, loyalty metrics, and primary purchase drivers.

- *Business Intelligence & Visualization (Power BI)*: Designing and developing an interactive dashboard that highlights critical patterns and trends, empowering stakeholders to make informed, data-driven decisions.
- *Actionable Insights & Recommendations*: Synthesizing key analytical findings and strategic business recommendations through a professional project report and a visual presentation tailored for executive leadership.
- *Version Control (GitHub Repository)*: Packaging all Python scripts, SQL queries, and Dashboard files into a well-structured, transparent, and reproducible repository architecture.

4. Key Analytical Dimensions

Based on the business requirements, this project delves into the following core metrics and segments:

- *Demographic Segmentation*: Revenue contribution analyzed by age group, gender, and geographical location.
- *Channel Performance*: Comparative analysis of sales performance and conversion rates between online platforms and physical offline stores.
- *Purchase Drivers*: Measuring the quantitative impact of promotional discounts, customer review ratings, and seasonality on overall spending behavior.
- *Customer Retention (Cohort Analysis)*: Tracking subscription conversion rates and retention metrics among repeat buyer cohorts.